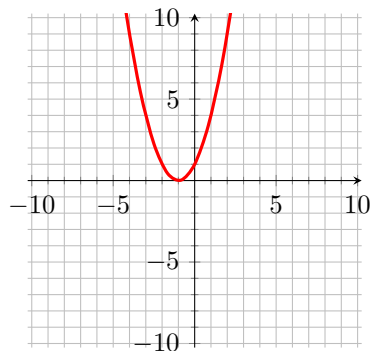


The graph of a quadratic function is given. Determine the function's equation from a set of options.

1. Check the direction, a .
2. Check the y -int.
3. Check the vertex.

Example. Determine the function's equation.



- | | |
|--------------------|----------------|
| (A) $x^2 + 2x + 1$ | (C) $x^2 - 1$ |
| (B) $x^2 - 2x + 1$ | (D) $-x^2 - 1$ |

We write quadratic functions in one of two ways; $f(x) = a(x - h)^2 + k$ or $f(x) = ax^2 + bx + c$.

$$f(x) = a(x - h)^2 + k$$

- Vertex: (h, k)
- $a < 0 \Rightarrow$ points down
- $a > 0 \Rightarrow$ points up
- x -int: $a(h)^2 + k$

$$f(x) = ax^2 + bx + c$$

- Vertex: $(-b/2a, f(-b/2a))$
- $a < 0 \Rightarrow$ points down
- $a > 0 \Rightarrow$ points up
- x -int: $f(0) = c$

Find the vertex/max/min of a quadratic function.

$$f(x) = a(x - h)^2 + k$$

- Vertex: (h, k)
- Max is k if it points down
- Min is k if it points up

Example. $2(x + 3)^2 - 10$.
 Vertex: $(-3, -10)$. Min: -10

$$f(x) = ax^2 + bx + c$$

- Vertex: $(-b/2a, f(-b/2a))$
- Max is $f(-b/2a)$ if it points down
- Min is $f(-b/2a)$ if it points up

Example. $2x^2 + 12x - 8$.
 $-12/2(2) = -3$ and $f(-3) = 2(-3)^2 + 12(-3) - 8 = -10$.
 Vertex: $(-3, -10)$. Min: -10

Determine whether the function is polynomial function.

A polynomial has the form: $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$.

- The graph of a polynomial is *smooth* and *continuous*.
- All the coefficients $(a_n, a_{n-1}, \dots, a_0)$ are REAL numbers. Real numbers include fractions and irrational numbers.
- All the exponents are POSITIVE INTEGERS; this also means that there are no x 's in the denominators.

ex. $f(x) = \sqrt{5}x^5 - 2x^2 + \pi$; YES, $\sqrt{5}$ and π are real numbers.

ex. $f(x) = x^5 - x^{3/2} + 2$; NO, $3/2$ is not an integer.

Use the Leading Coefficient Test to determine the end behavior of the polynomial function.

	a_n is positive	a_n is negative
n is even	$\nwarrow \cdots \nearrow$	$\swarrow \cdots \searrow$
n is odd	$\swarrow \cdots \nearrow$	$\nwarrow \cdots \searrow$

- n is the largest exponent, and a_n is the coefficient next to the x with the largest exponent.

ex. $f(x) = -x^4 + 2x^3 - 40x^2 + 1$. $a = -1$ and $n = 4$ so
 $f(x)$ falls to the left and falls to the right.

ex.

ex. $f(x) = 4x^2 + 2x^7 - 5x + 1$. $a = 2$ and $n = 7$ so
 $f(x)$ falls to the left and rises to the right.

Find the x -intercepts of the polynomial function. State whether the graph crosses the x -axis, or touches the x -axis and turns around.

- To find when the x -intercepts solve $f(x) = 0$.
- If a zero has an *even degree* then it turns around.
- If a zero has an *odd degree* then it crosses.

ex. $f(x) = x^3 - 3x^2$

- Solve $f(x) = 0$ by factoring $f(x) = x^2(x - 3) = 0$. So then the zeros are $x = 0$ and $x = 3$.
 - $x = 0$ has degree two; even $\Rightarrow$ it turns around.
 - $x = 3$ has degree one; odd $\Rightarrow$ it crosses.

ex. $f(x) = -2(x - 1)^2(x + 2)$

- $f(x)$ has zeros at $x = 1$ and $x = -2$.
 - $x = 1$ has degree two; even $\Rightarrow$ it turns around at 1.
 - $x = -2$ has degree one; odd $\Rightarrow$ it crosses at -2 .

Graph the above examples using the intercepts and end behavior.

- $f(x) = x^3 - 3x^2$

